

CPTG Redshift Stress Test of MQN01 J004131.9-493704

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Abstract

MQN01 J004131.9-493704, commonly known as the Red Potato, is a massive quiescent galaxy observed at spectroscopic redshift $z = 3.2496$ in the MQN01 cosmic-web structure. Published measurements describe a compact stellar body with integrated stellar velocity dispersion $\sigma_{\star} = 268 \pm 20 \text{ km s}^{-1}$, stellar mass of order $10^{11} M_{\odot}$, and rest-frame optical half-light radius near 0.9 kpc. This article uses the object as a high-redshift CPTG redshift stress test by comparing its local effective-radius support scale with the corresponding expansion-normalized acceleration scale. For the adopted midpoint radius, the stellar support scale is $2.5863 \times 10^{-9} \text{ m s}^{-2}$, which reduces to $5.3099 \times 10^{-10} \text{ m s}^{-2}$ after normalization by the expansion factor at the observed epoch. The result places the Red Potato on a strong early-time acceleration branch while retaining a present-equivalent scale in the compact-galaxy CPTG comparison range. The surrounding gas reservoir and elevated circumgalactic turbulence are treated as environmental context rather than as relaxed stellar support, making the object a useful stress case for separating compact stellar dynamics from high-redshift gas-rich structure.

1 Motivation

The CPTG [2, 3] redshift stress test asks whether the compact stellar support scale of a high-redshift galaxy becomes more interpretable after normalization by the expansion factor at the epoch of observation. The target is useful for this purpose because it combines a compact quiescent stellar body, a published integrated stellar velocity dispersion, and a secure spectroscopic redshift.

The source paper identifies the target as a quiescent galaxy in a gas-rich cosmic-web node. It also reports extended cool circumgalactic emission around the object and discusses a dynamically active environment [1]. Those environmental data are important context, but they are not inserted into the stellar support calculation. The redshift stress test is anchored to the compact stellar body through

$$\sigma_{\star}, \quad R_c, \quad z. \tag{1}$$

The test is defined by three quantities:

$$a_\sigma(R_c) = \frac{\sigma_\star^2}{R_c}, \quad (2)$$

$$E(z) = \frac{H(z)}{H_0}, \quad (3)$$

and

$$a_{\sigma,z}(R_c) = \frac{a_\sigma(R_c)}{E(z)}. \quad (4)$$

The purpose is not to construct a full radial dynamical solution. The purpose is to reduce the observed compactness and stellar dispersion to a redshift-aware acceleration scale, then tie that scale to the cosmic time at which the galaxy is observed.

In this article, the compact effective-radius scan variable is written as

$$R_c, \quad (5)$$

while the CPTG curvature radius is written as

$$R_C. \quad (6)$$

The two symbols should not be read as the same quantity.

2 Observed Inputs

The target is MQN01 J004131.9-493704, the quiescent galaxy commonly referred to as the Red Potato. The source paper reports a spectroscopic redshift of

$$z = 3.2496 \pm 0.0006, \quad (7)$$

where the uncertainty is quoted at the two-sigma level [1]. It also reports an integrated stellar velocity dispersion of

$$\sigma_\star = 268 \pm 20 \text{ km s}^{-1}, \quad (8)$$

and compact half-light radii along the major axis of approximately

$$R_c = 1.0 \text{ kpc} \quad (9)$$

and

$$R_c = 0.8 \text{ kpc}, \quad (10)$$

in the bands used for the compact morphology measurement [1]. The midpoint value used for the central calculation is

$$R_c = 0.9 \text{ kpc.} \quad (11)$$

The source paper describes the galaxy as massive and passive, with a stellar mass of order

$$M_\star \simeq 10^{11} M_\odot. \quad (12)$$

This mass statement is recorded as source context. The numerical redshift stress test below does not require the stellar mass; it depends only on the measured dispersion, compact radius, and redshift expansion factor.

Quantity	Adopted value	Role in test
Formal target	MQN01 J004131.9-493704	Object name
Redshift	3.2496	Expansion epoch
Stellar dispersion	$268 \pm 20 \text{ km s}^{-1}$	Support scale
Radius bracket	0.8–1.0 kpc	Size range
Midpoint radius	0.9 kpc	Central calculation
Stellar mass context	$\sim 10^{11} M_\odot$	Source context only
CGM extent	about 80 kpc	Environment

Table 1: Inputs used for the CPTG redshift stress test.

3 Local Stellar Support Scale

The local support scale is computed from the integrated stellar velocity dispersion and the adopted compact effective-radius variable:

$$a_\sigma(R_c) = \frac{\sigma_\star^2}{R_c}. \quad (13)$$

For the midpoint radius,

$$R_c = 0.9 \text{ kpc,} \quad (14)$$

or

$$R_c = 2.7771 \times 10^{19} \text{ m,} \quad (15)$$

and

$$\sigma_\star = 268 \text{ km s}^{-1} = 2.68 \times 10^5 \text{ m s}^{-1}. \quad (16)$$

Therefore,

$$a_\sigma = \frac{(2.68 \times 10^5)^2}{2.7771 \times 10^{19}} = 2.5863 \times 10^{-9} \text{ m s}^{-2}. \quad (17)$$

The dispersion uncertainty gives the midpoint-radius bracket

$$a_\sigma(248 \text{ km s}^{-1}) = 2.2147 \times 10^{-9} \text{ m s}^{-2}, \quad (18)$$

and

$$a_\sigma(288 \text{ km s}^{-1}) = 2.9867 \times 10^{-9} \text{ m s}^{-2}. \quad (19)$$

The compact stellar body therefore has a high local support scale before the redshift normalization is applied.

4 Expansion Factor and Cosmic Time

The redshift stress test uses the flat reference expansion factor

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \quad (20)$$

With

$$\Omega_m = 0.3, \quad (21)$$

$$\Omega_\Lambda = 0.7, \quad (22)$$

and

$$z = 3.2496, \quad (23)$$

the expansion factor is

$$E(z) = 4.8706. \quad (24)$$

For

$$H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (25)$$

the expansion rate at the observed redshift is

$$H(z) = H_0 E(z) = 340.95 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (26)$$

The scale factor at emission is

$$a_{\text{cos}} = \frac{1}{1+z} = 0.2353. \quad (27)$$

For the same reference expansion law, the cosmic age at redshift z is

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1} \left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}} (1+z)^{-3/2} \right). \quad (28)$$

This gives

$$t(z = 3.2496) = 1.93 \text{ Gyr.} \quad (29)$$

The corresponding present reference age is

$$t_0 = 13.47 \text{ Gyr,} \quad (30)$$

and the lookback time is

$$t_{\text{lookback}} = 11.54 \text{ Gyr.} \quad (31)$$

Thus the target is observed when the Universe is approximately

$$\frac{t(z)}{t_0} = 0.143 \quad (32)$$

of its reference present age.

5 Time-Linked CPTG Acceleration Scale

The redshift stress test can be written as a time-linked acceleration branch. Define the present-equivalent stress scale as

$$a_{\star,0}(R_c) = a_{\sigma,z}(R_c) = \frac{a_{\sigma}(R_c)}{E(z)}. \quad (33)$$

The acceleration scale at the emission time is then

$$a_{\star}(t_{\text{em}}; R_c) = a_{\star,0}(R_c)E(z). \quad (34)$$

For this redshift stress test,

$$a_{\star}(t_{\text{em}}; R_c) = a_{\sigma}(R_c). \quad (35)$$

For the midpoint radius,

$$a_{\star,0} = 5.3099 \times 10^{-10} \text{ m s}^{-2}, \quad (36)$$

and

$$a_{\star}(t_{\text{em}}) = 2.5863 \times 10^{-9} \text{ m s}^{-2}. \quad (37)$$

The corresponding CPTG curvature radius is tied directly to the same acceleration scale:

$$R_C(t) = \frac{48\pi c^2}{a_{\star}(t)}. \quad (38)$$

Thus, the early-time branch has a larger acceleration scale and a smaller curvature radius, while the present-equivalent branch has a lower acceleration scale and a larger curvature radius.

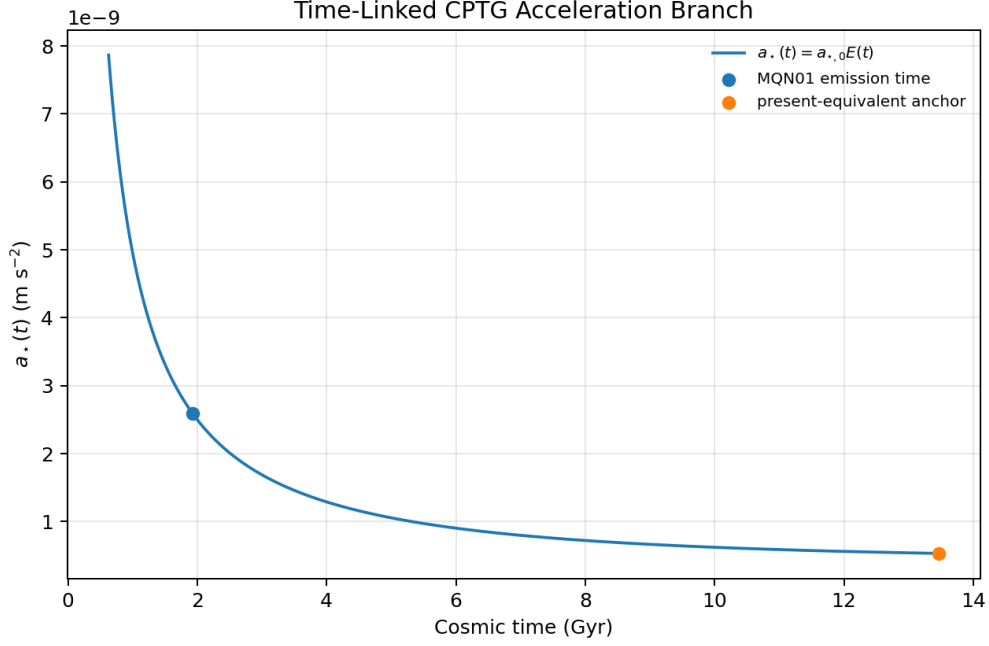


Figure 1: Time-linked CPTG acceleration branch for the midpoint radius. The curve uses $a_*(t) = a_{*,0}E(t)$, with $a_{*,0} = 5.3099 \times 10^{-10} \text{ m s}^{-2}$. The blue point marks the observed emission epoch at 1.93 billion years. The orange point marks the normalized present-equivalent anchor obtained by dividing the observed support scale by $E(z)$.

6 Radius-Scan Redshift Stress Scale

The normalized stress-test scale is

$$a_{\sigma,z}(R_c) = \frac{a_\sigma(R_c)}{E(z)}. \quad (39)$$

For the midpoint radius,

$$a_{\sigma,z} = \frac{2.5863 \times 10^{-9}}{4.8706} = 5.3099 \times 10^{-10} \text{ m s}^{-2}. \quad (40)$$

This value is the central numerical result of the redshift stress scale.

R_c (kpc)	a_σ (m s^{-2})	$a_{\sigma,z} = a_\sigma/E(z)$ (m s^{-2})	Comment
0.8	2.910×10^{-9}	5.974×10^{-10}	compact edge
0.9	2.586×10^{-9}	5.310×10^{-10}	midpoint
1.0	2.328×10^{-9}	4.779×10^{-10}	extended edge

Table 2: Redshift stress-test scale across the adopted radius bracket. The stellar dispersion is fixed at 268 km s^{-1} , and $E(z) = 4.8706$.

The same quantity can be displayed as a continuous function of the compact effective-radius variable,

$$R_c. \quad (41)$$

For this plot, the radius-dependent curve is

$$a_{\sigma,z}(R_c) = \frac{\sigma_\star^2}{R_c E(z)}. \quad (42)$$

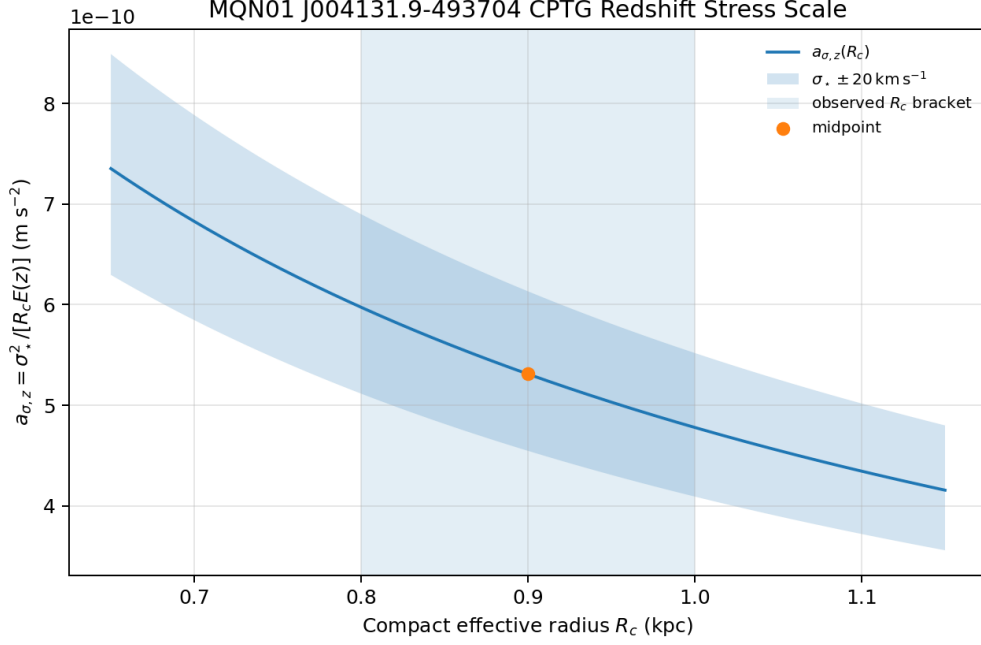


Figure 2: CPTG redshift stress-test scale as a function of compact effective radius. The curve shows the expansion-normalized support scale inferred from the measured stellar dispersion, while the surrounding band propagates the published dispersion uncertainty. The shaded vertical region marks the adopted compact-radius interval, and the orange point gives the midpoint value used for the article’s reference calculation.

The plot shows that the redshift-normalized stress scale remains within a narrow range across the adopted compact-radius interval. Increasing R_c lowers the stress-test scale because the dispersion support is divided by a larger effective radius.

7 Environmental Context

The redshift stress test separates the compact stellar body from the surrounding gas reservoir. The source paper reports extended Ly α , H α , and O III emission around the target, with the Ly α halo spanning roughly

$$80 \text{ kpc} \quad (43)$$

in diameter [1]. It also reports high circumgalactic line widths, including Ly α velocity dispersions of approximately

$$400\text{--}500 \text{ km s}^{-1} \quad (44)$$

at projected distances of roughly

$$20\text{--}30 \text{ kpc} \quad (45)$$

from the galaxy [1]. The source paper further notes that H α kinematics within the galaxy show no significant velocity gradient and are consistent with a dispersion-dominated gas state [1].

For the CPTG redshift stress test, these environmental quantities are not inserted into the stellar support scale. The stellar calculation is anchored to

$$\sigma_{\star}, \quad R_c, \quad z, \quad (46)$$

while the gas reservoir is treated as contextual evidence that the larger environment is dynamically active. This distinction keeps the test focused on the compact stellar body and prevents the turbulent circumgalactic medium from being misread as stellar pressure support.

8 Interpretation

At the midpoint compact radius, the local support scale is

$$a_{\sigma} = 2.5863 \times 10^{-9} \text{ m s}^{-2}. \quad (47)$$

After normalization by the expansion factor at

$$z = 3.2496, \quad (48)$$

the redshift stress-test scale becomes

$$a_{\sigma,z} = 5.3099 \times 10^{-10} \text{ m s}^{-2}. \quad (49)$$

The redshift corresponds to

$$t_{\text{em}} = 1.93 \text{ Gyr} \quad (50)$$

after the Big Bang in the reference cosmology. This places the stress test at an early cosmic epoch, when

$$H(z) = 340.95 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (51)$$

The time-linked acceleration relation can be summarized as

$$a_{\star}(t_{\text{em}}) = E(z)a_{\star,0}. \quad (52)$$

For the midpoint radius,

$$a_{\star,0} = 5.3099 \times 10^{-10} \text{ m s}^{-2}, \quad (53)$$

and

$$a_{\star}(t_{\text{em}}) = 2.5863 \times 10^{-9} \text{ m s}^{-2}. \quad (54)$$

Across the radius bracket, the expansion-normalized stress scale remains in the interval

$$a_{\sigma,z} \approx (4.8\text{--}6.0) \times 10^{-10} \text{ m s}^{-2}. \quad (55)$$

The key result is therefore a time-linked acceleration branch: the object is observed at an early cosmic time, its compact stellar support implies a high local acceleration scale, and dividing by $E(z)$ maps that scale to a present-equivalent value in the $10^{-10} \text{ m s}^{-2}$ range. The branch can also be read as a structural-clock sequence: high $a_{\star}$ at early time, declining $a_{\star}$ with cosmic expansion, and an expanding R_C toward the present-equivalent state.

9 Data Needed for a Sharper Redshift Stress Test

The current calculation is limited by the fact that several quantities are published in compact form. The redshift stress test would be sharpened by a posterior table containing a clearly stated radius convention,

$$R_{c,\text{maj}} \quad \text{or} \quad R_{c,\text{circ}}, \quad (56)$$

structural parameters,

$$q, \quad n, \quad (57)$$

and an aperture statement for the integrated stellar dispersion,

$$\sigma_{\star} = 268 \pm 20 \text{ km s}^{-1}. \quad (58)$$

These refinements would reduce systematic uncertainty in

$$a_{\sigma}, \quad \frac{a_{\sigma}}{E(z)}, \quad \text{and} \quad a_{\star}(t_{\text{em}}). \quad (59)$$

They are not required to define the present redshift stress test, but they would make the numerical result more precise.

10 Technical Summary

The target provides a focused CPTG redshift stress test at

$$z = 3.2496. \quad (60)$$

Using the published integrated stellar dispersion and the adopted midpoint radius gives

$$a_\sigma = 2.5863 \times 10^{-9} \text{ m s}^{-2}. \quad (61)$$

Using the expansion factor

$$E(z) = 4.8706, \quad (62)$$

the redshift-normalized result is

$$a_{\sigma,z} = 5.3099 \times 10^{-10} \text{ m s}^{-2}. \quad (63)$$

The observed redshift corresponds to

$$t_{\text{em}} = 1.93 \text{ Gyr}, \quad (64)$$

or a lookback time of

$$t_{\text{lookback}} = 11.54 \text{ Gyr}, \quad (65)$$

in the reference cosmology. The time-linked CPTG acceleration scale is

$$a_\star(t_{\text{em}}) = 2.5863 \times 10^{-9} \text{ m s}^{-2}, \quad (66)$$

with present-equivalent value

$$a_{\star,0} = 5.3099 \times 10^{-10} \text{ m s}^{-2}. \quad (67)$$

Across the radius bracket

$$R_c = 0.8\text{--}1.0 \text{ kpc}, \quad (68)$$

the redshift-normalized scale remains in the interval

$$a_{\sigma,z} \approx (4.8\text{--}6.0) \times 10^{-10} \text{ m s}^{-2}. \quad (69)$$

This is the main outcome of the article: the target supplies a compact, high-redshift, time-linked acceleration branch for a CPTG redshift stress test.

11 The Life Branch of the “Red Potato”

The galaxy is observed as it was when the Universe was only about 1.93 billion years old. Even at that early time, it was already massive, compact, dynamically hot, and largely quiet. That means we are not seeing the beginning of its life. We are seeing it after its first major transformation had already happened.

A reasonable birth window places the start of its main stellar assembly within the first billion years after the Big Bang, probably around 920 million years after the beginning of cosmic time, or earlier. This is not meant to be an exact birth date. It is a timing estimate based on the fact that the system already looked mature when the Universe was less than 2 billion years old. Its stellar population had enough time to build older stars, enrich itself, and settle into a compact body before the light we observe was emitted.

Its birthplace was a dense cosmic-web node. This was not an isolated, quiet region. It was an active early environment containing gas, galaxies, and energetic activity nearby. In such a place, gas could move along the cosmic web and feed rapid growth. The galaxy appears to have used that early supply efficiently, building a large stellar mass quickly while remaining very compact.

The unusual part of the story is that the galaxy became quiet while gas was still present around it. The environment did not simply run out of fuel. Instead, the available gas appears to have been difficult to use. Nearby activity and turbulent circumgalactic motion may have kept the gas from cooling, settling, and feeding new star formation. In that picture, the galaxy did not become quiet because the cosmic web was empty. It became quiet because the surrounding fuel was stirred, heated, or dynamically prevented from reaching the compact stellar body in a useful form.

In CPTG terms, this early compact phase corresponds to a strong local acceleration branch. Near the likely birth window, the branch was roughly twice as high as it was at the observed epoch. By the time the galaxy emitted the light we now see, the branch had already relaxed, but it was still much stronger than the present-equivalent value. This makes the acceleration scale useful as a kind of structural clock. It marks how compact and dynamically intense the system was at each stage of cosmic time.

The time sequence is simple. The galaxy likely began forming before the Universe was 1 billion years old. About 1 billion years later, when the Universe was 1.93 billion years old, it was already a compact quiescent system inside a gas-rich but unsettled environment. From that observed state to the present day, more than 11 billion years pass. Across that long interval, the smooth CPTG time branch lowers the characteristic acceleration scale by nearly a factor of five.

As that acceleration scale falls, the associated curvature radius grows. Early on, the system belongs to a stronger and more compact branch. Later, the branch relaxes toward a lower acceleration scale and a larger curvature radius. This

does not by itself describe every merger, disturbance, or size-growth episode the descendant galaxy may experience. Those would be additional structural changes. The redshift stress test tells the smoother background story: a compact galaxy born early, assembled quickly, quenched while still surrounded by gas, and then carried forward through cosmic time along a declining acceleration branch.

The result is a life picture rather than a single calculation. The galaxy begins in a dense early web, grows rapidly, becomes quiet surprisingly early, and is observed after about 1 billion years of development. Its later history is not directly observed here, but the time-scaled CPTG branch gives a clear direction: high early acceleration and compact structure give way, over billions of years, to lower acceleration and late-time relaxation.

References

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